Module 2: Intelligent Data Driven Applications on the Cloud

From notebooks to production-ready, AI-enabled applications

INTELLIGENT DATA DEVELOPMENT, JUNE 2026

ABOUT ME

LAURA COZMA

DATA SCIENCE DIRECTOR @PEPSICO

Education

- **Master in Data Science,** BSE - 2015
- **Master in Economic Analysis**, UAB - 2010
- **Bachelor in Economics - Cybernetics, Statistics and Business Information Systems**, Alexandru Ioan Cuza University, Romania - 2008 (Major: Computer Science. Minor: Economics)

Specialties

- **Data Science**
- **Machine Learning**
- **Data Engineering**
- **Cloud Architecture** (AWS Solutions Architect)

Kaggle ML Competitions Expert

Current Focus

Building and scaling production-grade forecasting and AI/data science systems in enterprise environments.

Experience

Data Science Director, PepsiCo (12.2022 – present)

- In charge of leading the IBP Data Science Europe Deployment
- **Keywords**: Azure Cloud, Databricks, Pyspark

Senior Manager, PepsiCo (06.2021 –12.2022)

- In charge of leading the IBP data science Iberia pilot
- **Keywords**: Azure Data Factory, Databricks, PowerBI, Pyspark

Data Strategy Lead, Ailylabs (02-05.2021)

- In charge of leading the data team and building up the AILY data strategy vision
- **Keywords**: AWS: Redshift, Postgres, EC2, Lambda, API Gateway, CloudFormation; Python

Analyst – Manager, Accenture (09.2015-01.2021)

- **Palantir and AWS Data Integration** @ Aerospace Company
- **Keywords**: AWS: EC2, Lambda, API Gateway, CloudFormation; Palantir: Code Authoring, Phonograph; Python, Pyspark

- **Foundry Slate Apps Industrialization** @ Aerospace Company
- **Keywords**: Palantir: Code Authoring, Slate, Phonograph; Python, Pyspark

- **Connected Factory** @ Manufacturer
- **Keywords:** AWS: Lambda, S3, IoT, Kinesis, Redshift, SageMaker, Python

WHY IS THE IDD COURSE SO IMPORTANT

This course gives you the foundations to grow from "doing data science" to building intelligent, AI-enabled applications.

When I started my career in 2015, I wish I had been exposed earlier to concepts like Git, Cloud, software engineering, deployment, and production-ready data science. At that time, many of these practices were still emerging in our field.

Today, the acceleration is even stronger. With the rise of Generative AI, copilots, and intelligent agents, **data scientists are expected not only to build models**, but to **design systems that use data and AI to solve real problems end to end**.

IDD is important because it connects the dots: data, engineering, cloud, AI, and product thinking.

THE INDUSTRY SHIFT - FROM DATA SCIENCE TO AI-ENABLED APPLICATIONS

AI accelerates development, but it also increases the need for engineering discipline.

Then	Now	Next
Notebook analysis	Production ML systems	AI-enabled applications
Manual experiments	Automated pipelines	Copilots and Agents
Local models	Cloud deployment	Human-in-the-loop systems
One-off outputs	Monitored products	Continuous intelligence

AI changes:

- How fast we can generate code
- How fast we can prototype
- How easily we can access explanations
- How users interact with data products
- How applications can automate workflows

AI does not remove the need for:

- Clear requirements
- Good data quality
- Version control
- Testing
- Cloud architecture
- Monitoring
- Responsible deployment
- Human judgment

WHAT IS THIS COURSE ABOUT

This course is about giving you the building blocks to move from idea to production. **A real intelligent data application needs more than an ML model.**

PRODUCT THINKING

What problem are we solving? For whom? How will success be measured?

AGILE DELIVERY

How do we organize work under uncertainty?

DEVOPS

How do we move from code to reliable software?

DATAOPS

How do we make data reliable, timely, and usable?

MLOPS

How do we train, deploy, monitor, and keep/improve models performance?

CLOUD

How do we run and scale the solution?

AI READINESS

How do we use AI safely and professionally?

MODULE MAP & EVALUATION

MODULE MAP

- **Part 1**: Data-driven Intelligence through DataOps and MLOps (class 1-2)
- **Part 2**: Cloud as a catalyst for data driven intelligence (class 3)
- **Part 3**: ***Hands on*** - Set up of cloud infrastructure for Forecasting (class 3 - 4)
- **Part 4**: Things change: Know why – Real Use Case Example @Pepsico (class 5)

MODULE EVALUATION

	Weight	**Owner**	**Evaluation mode**
Final project	60%	Javier + Laura	Team grade
Javier in-class exercises	20%	Javier	Individual / binary completion
Laura in-class exercises	20%	Laura	Team grade

AI POLICY

AI Policy
In-class exercises - AI tools are **not allowed.**

Final project
AI tools may be used as support, but students remain fully responsible for the final solution.

- AI usage must be disclosed.
- AI prompts / interaction history must be shared as part of the submission.
- AI-generated outputs must be reviewed, validated, and adapted by the students.
- No AI assistance is allowed during the video explanation or any follow-up clarification questions.
- Both students must speak in the video, with camera on.
- Professors may ask follow-up questions to validate understanding.

FINAL PROJECT - DEPLOYING A FORECASTING MODEL FOR GLOVO

- **Team**: Groups of 2 students. Groups of 3 allowed only by exception.
- **Due date**: **28th of June**
- **Evaluation**: The final project represents 60% of the module grade. The four project deliverables are equally weighted within the final project.
- **Project repository**: https://github.com/javiermas/bse-ts-public
- **AI policy:** AI allowed with disclosure, validation, and shared prompt history.

Project: The goal of the project is to deploy an AWS cloud solution that trains a model and generates predictions of orders using Glovo data.

Deliverables:
- **Code**
  - In any format, as long as it runs in the cloud. You may use any cloud-based compute environment of your choice.
- **Video (max 10 minutes)**
  - A walkthrough of your code and cloud setup. Ideally, you run the code on the cloud and generate an output file in the video. If your job takes longer than 10 minutes, you don't need to show the entire process—use caching where helpful. For recording the video, you may use any tool if you can share a link or file. If your tool limits video length, you can submit two clips.
- **Predictions** – A .csv file with out-of-sample predictions (format described below), stored in S3
- **Prod deployment considerations**
  - Section (PDF – max 5 pages) describing concepts thought in this part of the course: Define team(s) and their roles, interactions and ways of working. Define rough estimates for each part/ feature of the project and how these estimates were agreed. Define the architecture components of the end-to-end solution. Define data used and data flows (i.e. how data travels from the Storage/Database to your "forecasting tool" and connects to a visualization tool). Describe and argument environments to be used. Describe how the build and monitor of the ML Model will happen. Describe how the end-users will use your solution (visualization tool, interactive platform, etc.). Describe success KPIs.

FINAL PROJECT - DEPLOYING A FORECASTING MODEL FOR GLOVO CNT

Business requirements

You've just joined Glovo. The company uses a **slot-based system**, dividing each day into **24 hourly slots** per city. For efficient operations, Glovo needs to predict the optimal number of couriers required in each slot. Overestimating leads to idle couriers and wasted resources; underestimating causes delivery delays.

Currently, Operations manually determines these numbers based on past demand, but as Glovo expands, this manual process no longer scales. Your task is to help automate it by **forecasting hourly order volumes**.

To simplify, we'll focus on **one city only**. Every **Sunday at 23:59**, your model should forecast the **number of orders expected in each hourly slot** for the upcoming week (from **Monday 00:00 to Sunday 23:00**, inclusive). You may use all data up to and including the current week to generate your forecast.

EDA

- The data is under *data/train_data.csv.* Explore the data, visualize it. Look for trends, cycles and seasonality. Also, can you find any outliers? How does your EDA inform your modelling?

Modelling

- Try different models. Validate each model in a way that would imitate the real problem (every Sunday you forecast all next week). Watch out for data leakage. Evaluate each model on MSE and SMAPE. Which one performs better?

Output format

- A .csv file with columns: time (hourly resolution), datetime64[ns] type and preds: float64 type, 24*7 rows (7 days worth of predictions)

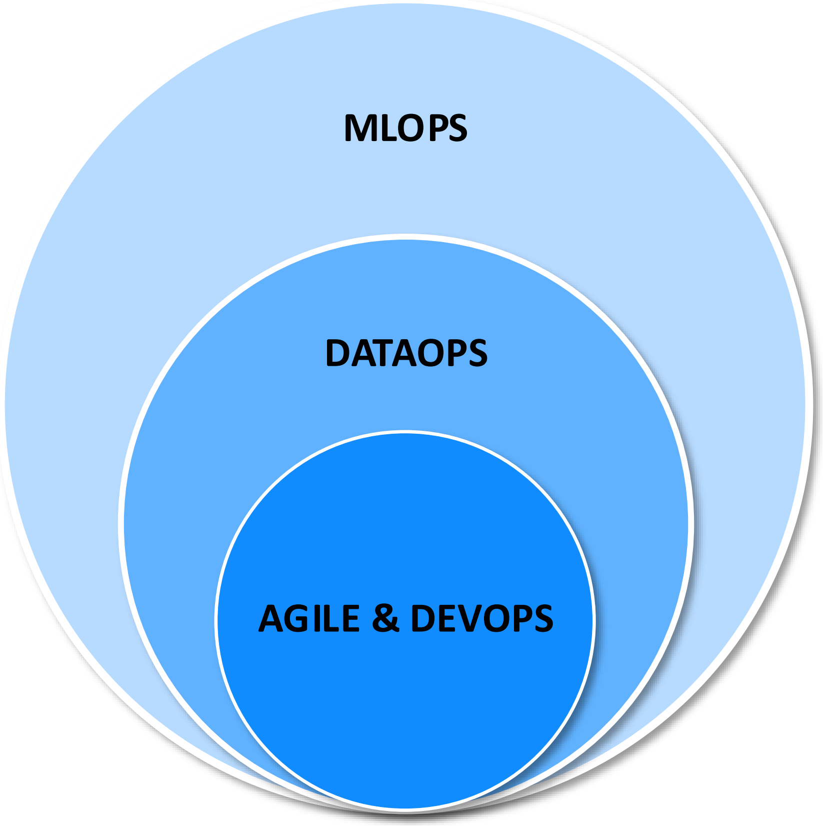
Going from **2022-01-24 00:00:00** to **2022-01-30 23:00:00** (both inclusive)

- You can check your output format with the function provided to you in the github repository. There might be some check we've missed, so make sure the output is what you want it to be.

Part 1: Data-driven Intelligence through DataOps & MLOps

INTELLIGENT DATA DEVELOPMENT, JUNE 2026

CONCEPTS COVERED – ZOOM IN



Traditionally, **machine learning** has been approached from a perspective of **individual scientific experiments predominantly carried out in isolation by data scientists**.

However, **as machine learning models become part of real-world solutions and critical to business**, we will have to **shift our perspective**, not to depreciate scientific principles but to make them more **easily accessible, reproducible and collaborative.**

We can achieve accessibility, reproducibility and collaboration in data science **through agile methodologies** - we iteratively improve flexibility and adaptability in project development and **through implementing DevOps** workflow - automated testing and continuous integration enhance collaboration and efficiency in teams.

AGILE PRODUCT DELIVERY AND DEVOPS

Any time we talk *about **best practices*** or **approaches to plan and deliver successful software**, you'll hear the terms agile and DevOps.

Agile and DevOps they both try to **fix very specific problems**, **problems encountered again and again using the software planning and development methods** that were typical from, let's say, the 1950s, all the way through the 1990s.

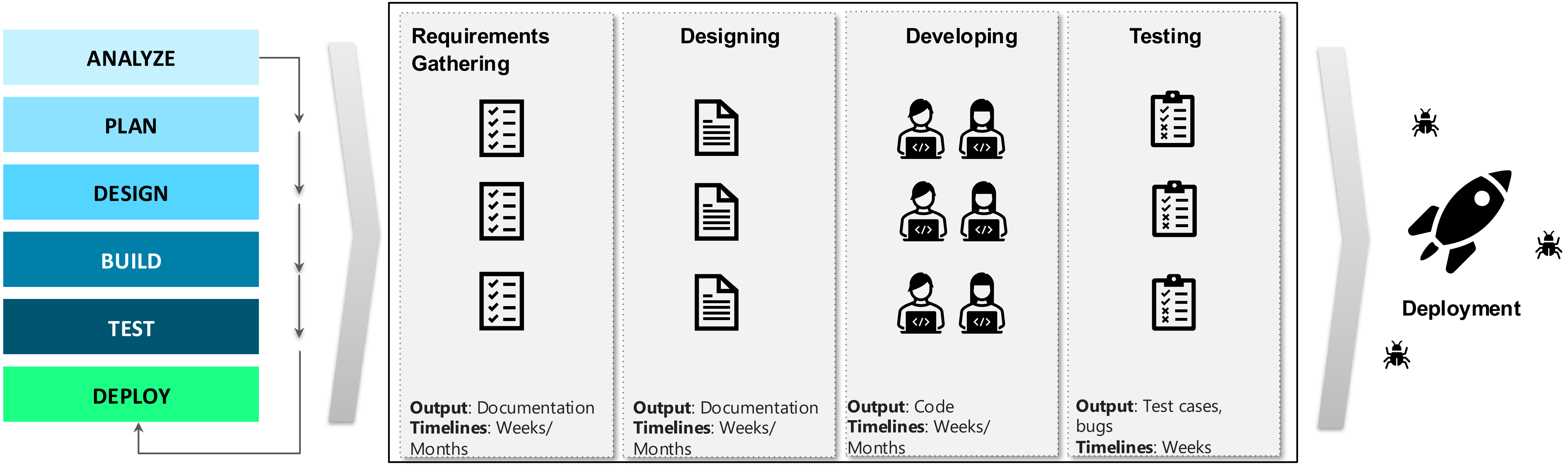
WHAT IS AGILE?

”Agile is **an iterative approach to project management and software development that helps teams deliver value to their customers faster and with fewer headaches**. Instead of betting everything on a "big bang" launch, an agile team delivers work in small, but consumable, increments.”

ATLASSIAN

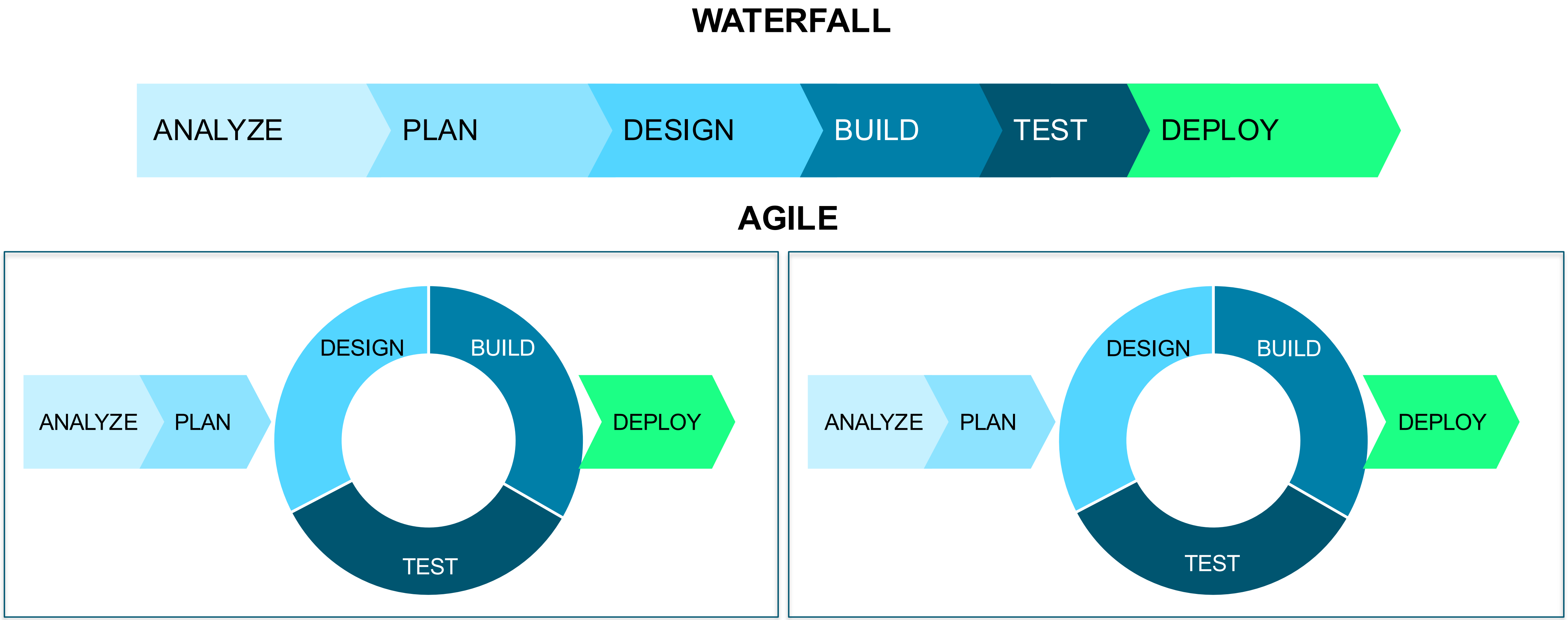
THE WATERFALL MODEL

- A **breakdown of project activities into a formalized highly structure linear sequential phases,** usually 5 or 6, where each phase depends on the deliverables of the previous one and corresponds to a specialization of tasks.
- **It allows for departmentalization and control**. A schedule can be set with **deadlines for each stage of development** and a product can proceed through the development process model phases one by one.
- From collecting requirements to the time the software was finally deployed, years had often passed. **It's entirely possible that the original requirements have changed or they weren't even properly understood.**



WATERFALL VS AGILE

Agile was developed as a response to Waterfall's more rigid structure. Given a software development project can take years to complete, technology can change significantly during that time. Agile was developed as a flexible method that welcomes **incorporating changes of direction even late in the process**, as well as accounting for **stakeholders' feedback throughout the process**.



AGILE VS WATERFALL AND WHY AGILE FITS DATA SCIENCE

Waterfall:

- **Linear, sequential model** with fixed phases (e.g., requirements → design → build → test)
- **Limited adaptability** once project starts
- Not ideal for evolving Data/DS/AI projects

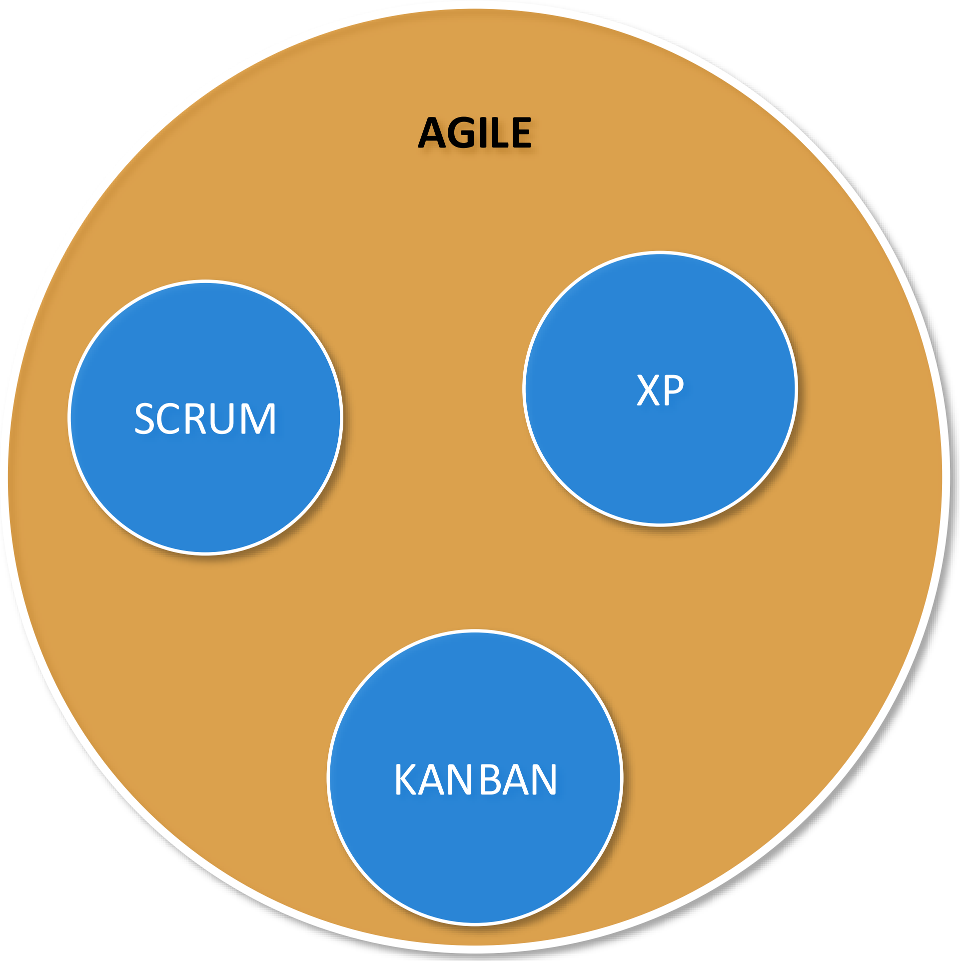
Agile:

- **Iterative and incremental** development
- **Embraces change** and continuous feedback
- Designed for **flexibility, collaboration, and faster delivery**

Why Agile Fits Data Science

- Data projects evolve with discoveries; Agile allows adapting to new insights quickly.
- DS projects are exploratory by nature and rarely have fixed outputs
- Agile allows quick iteration and fast feedback
- Helps prioritize features/models based on business value
- Encourages frequent stakeholder engagement
- Drives delivery of incremental insights, not just final models

AGILE FRAMEWORKS OVERVIEW



Agile is a **mindset.**

Scrum, Kanban, XP (Extreme Programming) are **Agile frameworks**.

Scrum and Kanban are the frameworks mostly used in data science projects.

Kanban methodology: Visual board, continuous delivery → best for support/sustain teams or evolving tasks

Scrum framework: Sprints, structured roles, ceremonies → best for well-defined project cycles

XP (Extreme Programming): Engineering-heavy, rarely used in DS

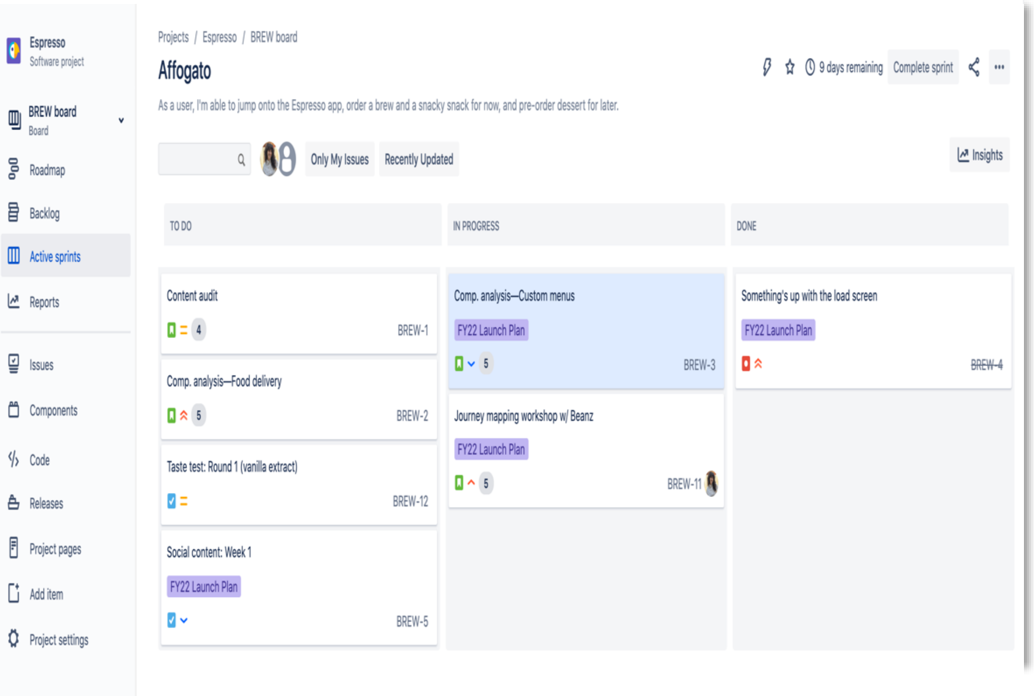
SCRUM – CORE CONCEPTS

Roles: Product Owner (PO), Scrum Master (SM), Team

Events: Sprint Planning, Daily Standup, Sprint Review, Retrospective

Artifacts: Product Backlog, Sprint Backlog, Increment
Sprints are time-boxed (1–4 weeks) with clear outcomes

Emphasis on team ownership, transparency, and "Definition of Done"



Scrum Board Example, Atlassian

SCRUM - ROLES

PRODUCT OWNER

- Defines and accepts user stories
- Acts as the customers for developer questions
- Works with Product Management to plan PIs (Product Increment)

SCRUM MASTER

- Coaches the agile team and facilitates team meetings
- Remove impediments and protects the team from outside influence
- Attends scrum of scrum meetings

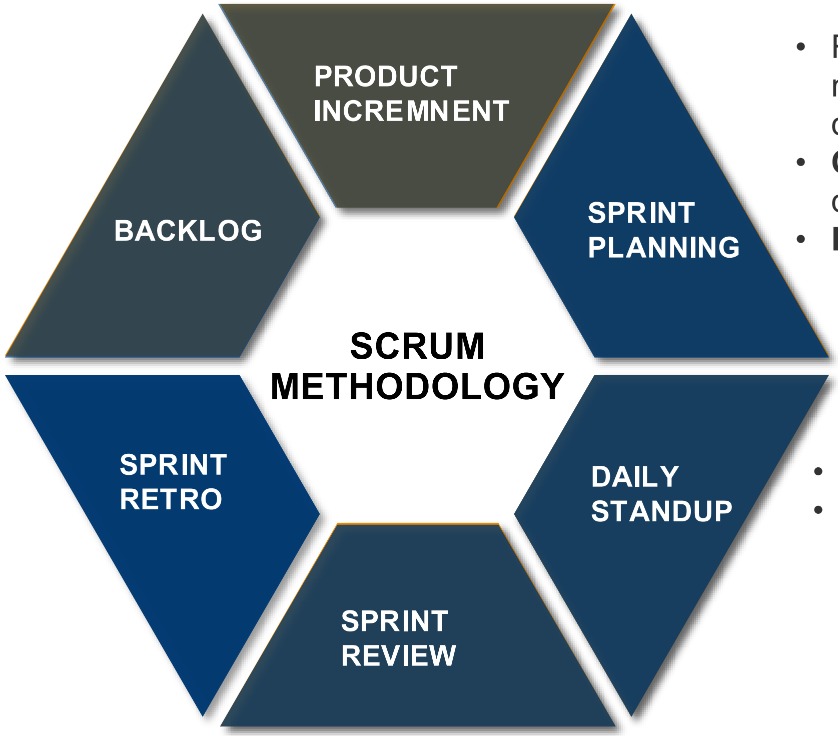
TEAM

- Create and refine user stories and acceptance criteria
- Define, build, test and deliver stories
- Develop the PI
- Objectives Iteration plans
- 5-11 members

SCRUM – EVENTS & ARTEFACTS

- Recursive fase-to-fase event where all teams work with the product owners on features and user stories they can commit
- Contains 5 sprints, last sprint is usally the innovation sprint
- **Frequence**: every 10 weeks, might vary

- The **Team Backlog** contains user and enabler stories that originate from the **Program Backlog**, as well as stories that arise locally from the team's local context.
- The Product Owner (PO) is responsible for the team backlog.



- Recursive event: team members determine how much of the team backlog they can commit to delivering.
- **Output**: List of stories committed with clearly defined acceptance criteria.
- **Frequence**: every 1-4 weeks

- Retrospective is used by Agile teams to reflect on the iteration just completed and to come up with ideas to improve team involvement and processes.
- Can have different formats: "what went well", "to improve", "Actions/Kudos"

- Daily meeting to assess team progress
- **Agenda**:
  - What did I do yesterday?
  - What is the plan for today?
  - Are there any blocking points to be addressed?

- The Sprint Review is an inspect and adapt point at the end of the Sprint.
- Business Owners, stakeholders assess what has been built during the sprint and discuss changes and new ideas.
- The teams, Product Owner and stakeholders decide the direction of the product.

SCRUM – DEFINITION OF DONE

DONE means is live in production and our customers can use it.

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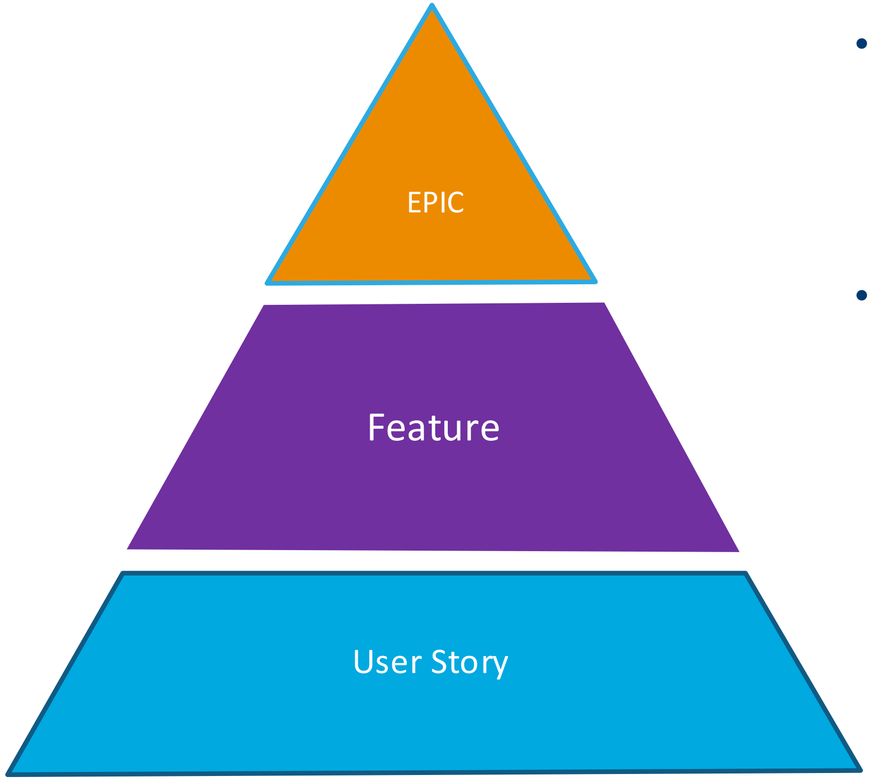
DONE means I have finished coding and it is ready to be tested.

SCRUM – HOW WE TRACK WORK



- In Agile, work is structured to *aid in the development & delivery of a product*.
  - They should not be defined/created to map to a business process. These are two different areas of focus when implementing a solution.

- The three most used elements are - **Epics**, **Features**, and **User Stories**.
  - **Epics** represent a major capability/scope of the product that can be broken down into small parts. Epics are commonly completed in **less than 3 – 6 months**.
  - **Features** represent a key aspect of the product and allows readers to understand clearly what part of the product is built. Feature are commonly completed in **less than 12 weeks**.
  - **User Stories** are a small building block of a Feature that clearly identifies a specific scope of work. A user story can be fully designed/built/tested **in less than a sprint.** It is usually owned by one single person with limited support needed by others.

KANBAN – CORE CONCEPTS

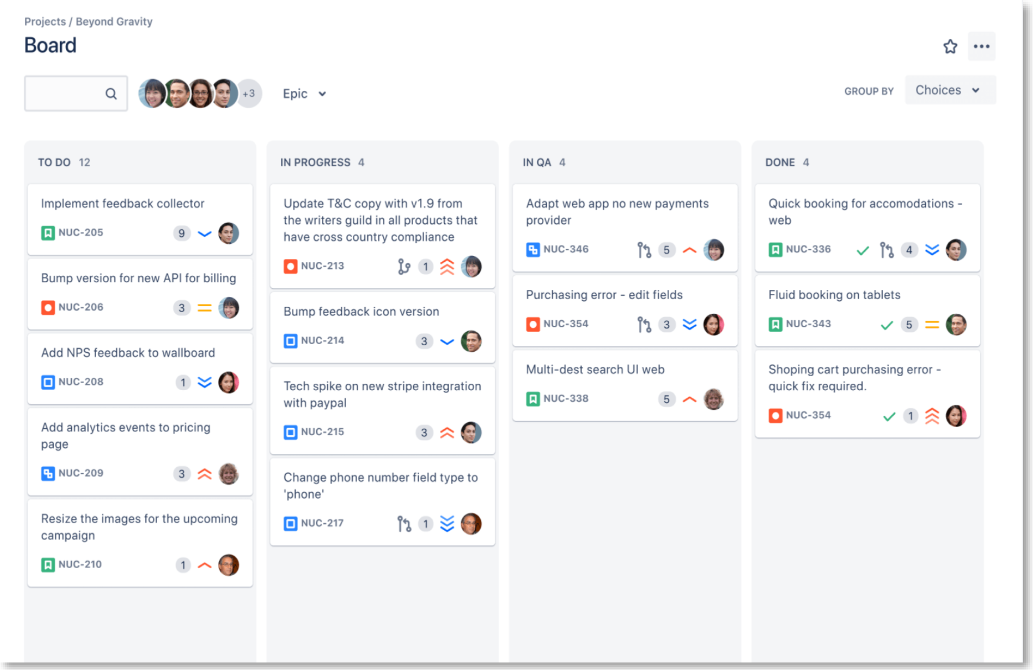
Visualize tasks as cards on a board (e.g., To Do → In Progress → Done)

Use WIP limits to prevent overloading team

Emphasize flow over iteration

No defined sprints; ideal for ongoing analysis, support, or experimentation

Pros: Simplicity, visibility, continuous delivery
Cons: Less structure, can become chaotic without discipline



Kanban Board Example, Atlassian

SCRUM/KANBAN - WORKING SESSION – 35' + 20' PRESENTATION

Goal: Simulate how a **data science team** would plan a forecasting project using the **Scrum/Kanban framework**.

Task - 35':

- **Scrum**: Each team (pair of 2) needs to create and populate their corresponding EPIC (a big goal or deliverable) and break it into FEATURES – functional blocks that can be delivered in 2-3 weeks. Split each feature into User Stories and estimate the amount of time needed. Use **story points** (1 = easy, 5 = complex) or **hours-** Multiply by 2 your time estimates. **Plan a Sprint (2-week cycle):** What subset of User Stories can the team commit to in one sprint?
- **Kanban:** Set up a Kanban board containing the below stages of work: *Backlog* (Tasks to be done), *To Do* (Tasks ready to start), *In Progress* (Tasks currently being worked on), Te*sting* (Tasks to be tested), *Done* (Completed tasks). Add task cards (should represent concrete DS activities) and set WIP limits. **Review flow and adjust**. Ask: Is anything stuck? Where's the bottleneck? Discuss how the team would coordinate to unblock flow
- **Share-out**: 5' per team

- ***Team 1*: Data acquisition, cleansing and preprocessing (EPIC)**
  - Ideas:
    - Explore where to find free API holiday and weather data (FEATURE)
    - Gather promo and price information from internal sources (commercial) (FEATURE)
    - Understand where to find sales data. (FEATURE)
    - Propose cleansing methodologies (handling missing values, outlier removal, remove/or not outliers that are known events as strikes, customer clashed etc) (FEATURE)
- ***Team 1*: Exploratory Data Analysis and Health Checks on data (EPIC)**
  - Ideas:
    - Perform analysis to understand correlation with sales for holidays, weather, promotion data and price(FEATURE)
    - Generate visualizations year-over-year to see whether Easter, Christmas, Black Friday etc. have impacts on sales (FEATURE)
    - Create a list of health checks for the data (i.e. data should have the defined structure, primary and foreign key enforcements, etc). (FEATURE)
- ***Team 2*: Feature Engineering (EPIC)**
  - Ideas:
    - Features for sales (lags, trend, seasonality, yearly average, total sales by category, rolling means, variances, data/time features etc) (FEATURE)
    - Features for holidays (lags, long weekends, holiday flag, etc) (FEATURE)
    - Features for promotions (cannibalization of most similar product, on promo flag, promo discount etc) (FEATURE)
    - Features for price (price flag, price increase of more than 50% etc) (FEATURE)
- ***Team 2*: Model Selection and Training (EPIC)**
  - Ideas:
    - Define what models will be used (time series, Machine Learning, Deep Learning etc)(FEATURE)
    - Hyperparameter tuning and feature selection. (FEATURE)
    - Training and Testing of each model (FEATURE)

- ***Team 3*: Data acquisition, cleansing and preprocessing**
  - Ideas:
    - Explore where to find free API holiday and weather data
    - Gather promo and price information from internal sources (commercial)
    - Understand where to find sales data.
    - Propose cleansing methodologies (handling missing values, outlier removal, remove/or not outliers that are known events as strikes, customer clashed etc)
- ***Team 3*: Exploratory Data Analysis and Health Checks on data**
  - Ideas:
    - Perform analysis to understand correlation with sales for holidays, weather, promotion data and price.
    - Generate visualizations year-over-year to see whether Easter, Christmas, Black Friday etc. have impacts on sales.
    - Create a list of health checks for the data (i.e. data should have the defined structure, primary and foreign key enforcements, etc).
- ***Team 4*: Feature Engineering**
  - Ideas:
    - Features for sales (lags, trend, seasonality, yearly average, total sales by category, rolling means, variances, data/time features etc)
    - Features for holidays (lags, long weekends, holiday flag, etc)
    - Features for promotions (cannibalization of most similar product, on promo flag, promo discount etc)
    - Features for price (price flag, price increase of more than 50% etc)
- ***Team 4*: Model Selection and Training**
  - Ideas:
    - Define what models will be used (time series, Machine Learning, Deep Learning etc).
    - Hyperparameter tuning and feature selection.
    - Training and Testing of each model.